

Schubert Cherry

Prunus virginiana 'Schubert'

Species background: Discovered growing wild along the Missouri River near Bismarck, North Dakota, and introduced as a cultivated variety by the Oscar H. Will Company nursery in 1943. It's one of the only street trees where a single tree can show both green and purple leaves at the same time, because new leaves emerge green and only turn purple as they mature through the season.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Schubert Cherry: Find both green and purple leaves on the same schubert cherry tree.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, green and purple crayons.

Teacher background

Look closely -- I might have both green and purple leaves on me at the same time! My new leaves start out green and slowly turn purple as summer goes on.

Procedure

- 1 Engage: Ask students if they've ever seen one plant with two different colored leaves.
- 2 Explore: Look for both green and purple leaves on the same tree.
- 3 Explain: Explain that new leaves start green and slowly change to purple.
- 4 Art: Find a green leaf and a purple leaf on the same tree and draw both.
- 5 Evaluate: Student can find and describe both leaf colors on one tree.

Discussion questions

Why might new leaves look different from older ones?

Assessment

Correctly identifies both green and purple leaves.

Extension

Watch one branch over several weeks to see if more leaves turn purple.

GRADE BAND 3-5

Schubert Cherry: Describe the green-to-purple color change as the leaf matures.

Standards connection: NGSS 4-LS1-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Some leaves start green and shift to purple as summer goes on -- try watching one leaf over a few weeks.

Procedure

- 1 Engage: Ask what students think causes a leaf to change color over its lifetime, not just in fall.
- 2 Explore: Compare a newer, greener leaf to an older, more purple one on the same branch.
- 3 Explain: Explain that this color shift happens as the leaf matures through the growing season, not in fall.
- 4 Art: Draw one branch that's half green leaves, half purple leaves, showing the tree changing right before your eyes.
- 5 Evaluate: Student explains in one sentence that newer leaves are greener and older leaves are more purple.

Discussion questions

What would you expect this tree to look like by the end of summer?

Assessment

Correct one-sentence explanation of the maturation color shift.

Extension

Sketch the same branch again in a few weeks if you can revisit it.

Schubert Cherry: Explain developmental pigment masking as the cause of the green-to-purple shift.

Standards connection: NGSS MS-LS1-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, materials for a 3-panel triptych.

Teacher background

New leaves start green because chlorophyll masks anthocyanin that's already forming; the balance shifts visibly as the season goes on.

Procedure

- 1 Engage: If a leaf has two pigments in it from the very start, why might it look like only one color at first?
- 2 Explore: Research how one pigment can visually mask another when both are present.
- 3 Explain: Explain chlorophyll masking anthocyanin, and how the balance shifts as the leaf matures.
- 4 Art: Create a time-lapse-style triptych -- 3 panels -- showing the same leaf transitioning from green to purple over the season, based on your own observations.
- 5 Evaluate: Write a paragraph explaining pigment masking using this species as the example.

Discussion questions

How is this different from red maple, where the red anthocyanin is made fresh in fall rather than being masked all along?

Assessment

Paragraph correctly explains pigment masking and distinguishes it from fresh pigment production.

Extension

Compare this masking process to red maple's active fall anthocyanin production.

Schubert Cherry: Design a data-driven study tracking a single leaf's pigment intensity over time.

Standards connection: NGSS HS-LS1-5; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, camera or color-sampling tool, notebook or spreadsheet.

Teacher background

New leaves start green because chlorophyll's green color masks anthocyanin pigments already being produced; research the biochemical process by which chlorophyll degrades faster than anthocyanin in this cultivar, unlike most trees, where anthocyanin appears only in fall.

Procedure

- 1 Engage: What would a graph of 'green-ness' versus 'purple-ness' over time look like for a single leaf on this tree?
- 2 Explore: Research the biochemistry of differential chlorophyll degradation in this cultivar.
- 3 Explain: Explain why this cultivar's pigment timeline is unusual compared to most trees.
- 4 Art: Design a data-driven color-change chart, photographing or sampling the same leaf weekly and plotting a rough color-intensity trend over several weeks.
- 5 Evaluate: Submit the biochemistry research summary and the data-driven color-change chart.

Discussion questions

What environmental or genetic factors might cause chlorophyll to degrade faster than anthocyanin in this cultivar specifically?

Assessment

Rubric covering biochemical research accuracy and quality of the data chart.

Extension

Compare this cultivar's pigment timeline to Crimson King maple's, which stays purple continuously rather than shifting.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.